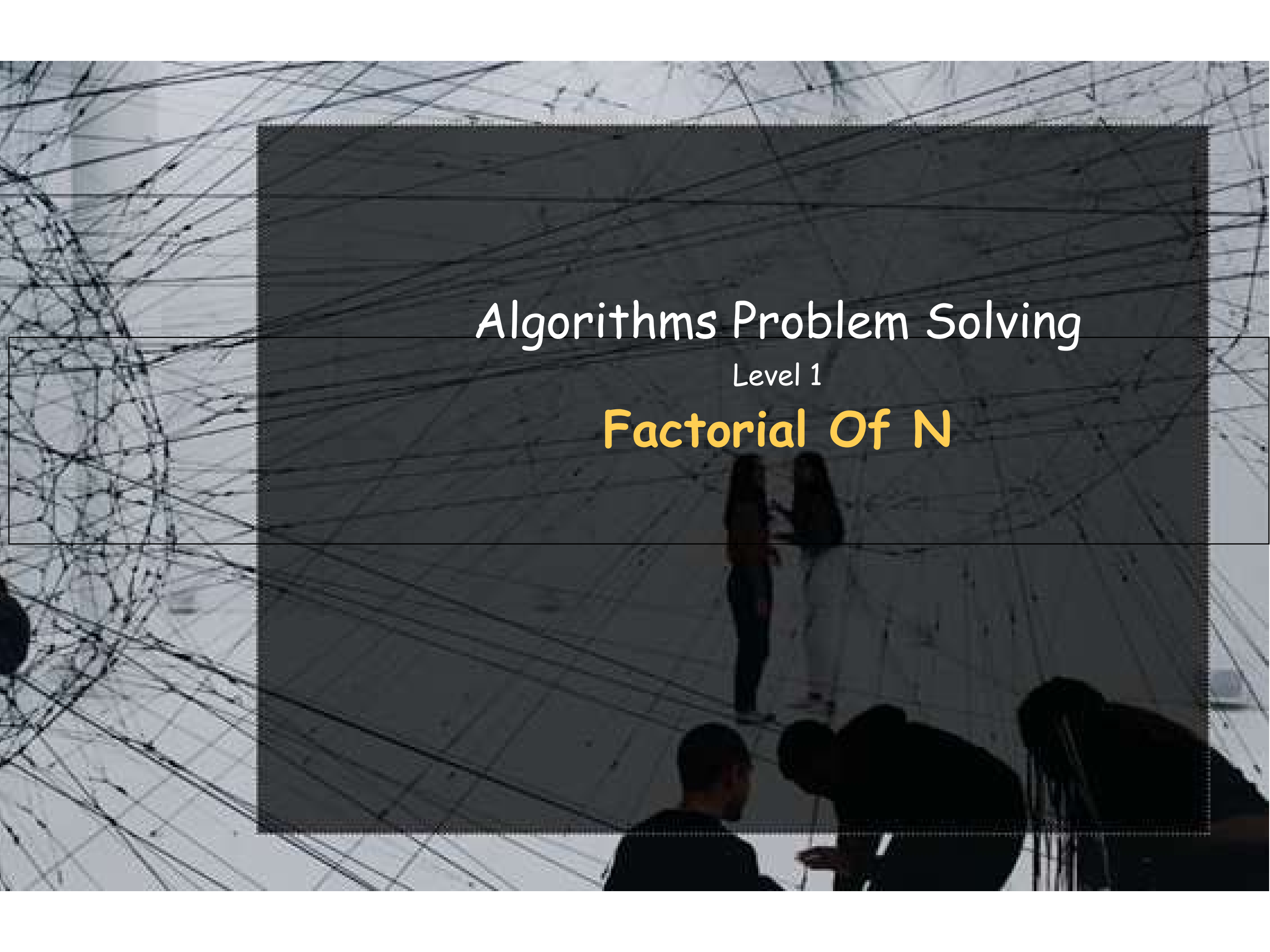


Algorithms Problem Solving

Level 1

Factorial Of N



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Definition

- ◆ This program reads a non-negative number N and calculates its factorial.
- ◆ The factorial of N is the product of all positive integers from 1 to N .
- ◆ It uses a `do-while` loop to make sure the user enters a non-negative number.
- ◆ It uses a `for` loop to calculate the factorial.

Explanation

- ◆ ``ReadPositivNum()`` repeatedly asks the user for a number until a non-negative value is entered.
- ◆ If the number is negative, an error message is displayed and the user must try again.
- ◆ ``Factorial()`` starts ``factorial`` at 1.
- ◆ The ``for`` loop starts at 2 and multiplies ``factorial`` by each number up to N.
- ◆ The multiplication operator ``*=`` is shorthand for:
``factorial = factorial * i``
- ◆ ``printResult()`` receives N and its factorial and prints the result.
- ◆ ``printResult(N, Factorial(N))`` calculates the factorial first, then passes the returned value to ``printResult()``.

Steps in main()

1. Read a number from the user.
2. Check if the number is negative.
3. If negative, ask the user to enter another number.
4. Set `factorial` to 1.
5. Start the loop from 2.
6. Multiply `factorial` by the current number.
7. Continue until reaching N.
8. Return the factorial.
9. Print the result.

💡 Example

◆ If the user enters:

$$N = 5$$

◆ The calculation is:

$$1 \times 2 \times 3 \times 4 \times 5 = 120$$

➡ Result

Factorial of 5 is 120

Note

- ◆ The factorial of 0 is 1:
`0! = 1`
- ◆ This is why `factorial` starts at 1 and the loop starts at 2.
- ◆ For example:
 - `1! = 1`
 - `2! = 2`
 - `3! = 6`
 - `4! = 24`
 - `5! = 120`

⊘ Warning

- ◆ The function returns ``int``, so large factorials can exceed the range of the ``int`` data type.
- ◆ For example, ``13!`` is already larger than the maximum value normally supported by a 32-bit signed ``int``.



Comparison

Factorial

- ✓ Uses multiplication.
- ✓ $5! = 1 \times 2 \times 3 \times 4 \times 5$.
- ✓ $0! = 1$.

Sum

- ✓ Uses addition.
- ✓ $1 + 2 + 3 + 4 + 5 = 15$.
- ✓ Does not represent factorial.

✦ Summary

- ✓ Factorial is the product of all positive integers from 1 to N.
- ✓ `0!` and `1!` are both 1.
- ✓ The `do-while` loop prevents negative input.
- ✓ The `for` loop calculates the factorial.
- ✓ `*=` multiplies the current factorial by the loop counter.
- ✓ `int` can overflow when calculating sufficiently large factorials.

Thank you 😊